Applied Convex Optimization

TAMU ECEN-629

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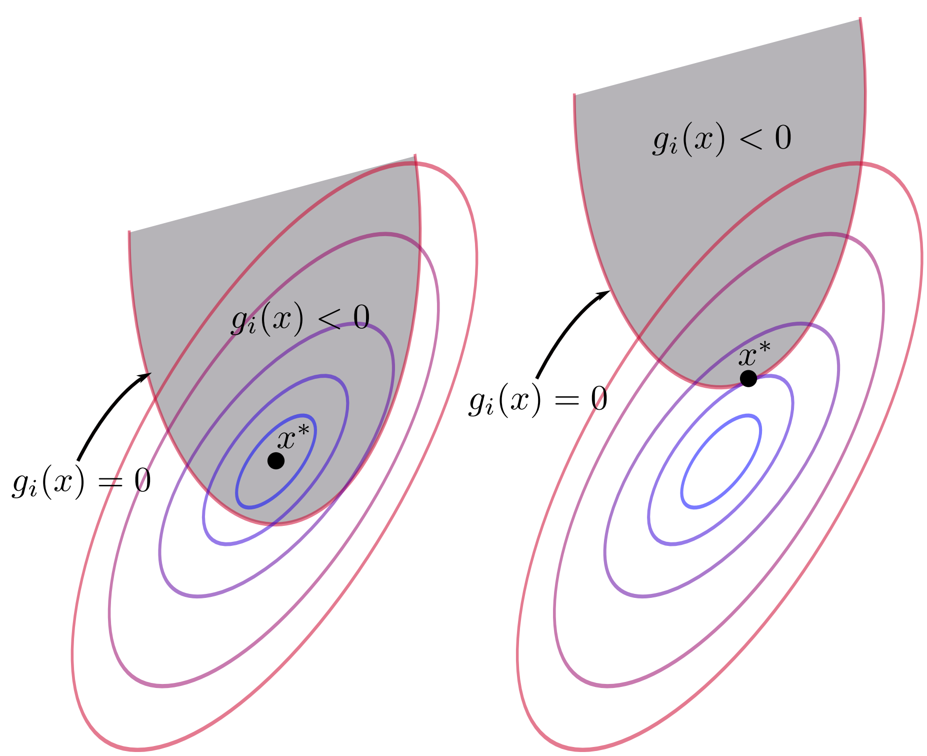
Fall 2025



Acknowledgements

These notes were compiled from the course materials for ECEN-629 (Applied Convex Optimization) taught by Dr. Chao Tian at Texas A&M University during the Fall 2025 semester.

Applied Convex Optimization (ECEN-629)

Instructor: Dr. Chao Tian
Semester: Fall 2025
Time: TTh 12:45pm - 2:00pm
Location: ETB 1003

Description: Introduction to convex optimization including convex set, convex functions, convex optimization problems, KKT conditions and duality, unconstrained optimization, and interior-point methods for constrained optimization; applications in information science, digital systems, networks and learning.

Contents

Chapter 1

Introduction

Optimization. Intuitively, we encounter these problems quite often in our lives.

$$\text{minimize } f(x) \text{ subject to } f_i(x) \leq b_i, \quad i = 1, \dots, n$$

where $x = (x_1, \dots, x_n)$ $f_0 : \mathbb{R}^n \to \mathbb{R}$ is the objective function $f_i : \mathbb{R}^n \to \mathbb{R}$ are the constraint functions

the optimal solution x^* gives the smallest value of f_0 that satisfies the constraints.

And exercise: aim to balance time spent studying for final exams.

$$\text{maximize } t_1\alpha_1 + t_2\alpha_2 + t_3\alpha_3 \text{ subject to } 10d \geq t_1 \geq 0 \dots t_1 + t_2 + t_3 = 14d$$

Optimization methods: least squares

$$\text{minimize } ||Ax - b||_2^2$$

has analytical solution

$$x^* = (A^T A)^{-1} A^T b$$

Optimization methods: linear programming

$$\text{minimize } c^T x$$

subject to

$$a_i^T x \leq b_i, \quad i = 1, \dots, m$$

has no analytical solution

Optimization methods: convex optimization

$$\text{minimize } f_0(x)$$

subject to

$$f_i(x) \leq b_i, \quad i = 1, \dots, m$$

has no analytical solution. often difficult to recognize.

Example: m lamps illuminating n patches intensity I_k at patch k depends on lamp powers p_j

$$I_k = \sum_{j=1}^{m} a_{kj} p_j \quad a_{kj}^{-2} \max\{\cos\theta_{kj}, 0\}$$

Want to minimize

$$|\log I_k - \log I_{\text{des}}|$$